OVERGROUND WALKING IN VR EVACUATION STUDIES: DESIGN CONSIDERATIONS AND REPRODUCIBILITY GUIDELINES

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1. INTRODUCTION

Virtual Reality (VR) has been widely adopted in Human Behaviour in Fire (HBiF) research to study evacuation behaviour and decision making, with growing interest in using VR to collect pedestrian movement, or locomotion, data. However, the ecological validity of locomotion data collected in VR remains relatively underexplored. Validity in this context has both internal dimensions (the extent to which an experiment can measure a genuine causal relationship) and external dimensions (generalisability of findings to contexts beyond the experimental setting); both are relevant for VR locomotion research (Haghani, 2023). Presently, there is no unified guidance to assist VR researchers in designing studies that aim to maximise the validity of movement data collected in VR.

This paper focuses on overground walking in VR, that is, natural, room-scale walking in which a user can freely move through the real environment (RE), with their physical position and orientation tracked and translated to their movement through a virtual environment (VE). We do not address alternative locomotion techniques such as controller-based locomotion, walking-in-place, redirected walking, or repositioning devices such as treadmills.

Studies consistently report systematic differences in spatio-temporal gait parameters between RE-VE. These differences are typically characterised by slower walking speeds, shorter step lengths, and increased base of support (wider step width and increased double support time) in VEs (Sato et al., 2026) compared to REs. The magnitude of this 'movement realism gap' varies considerably across studies (from negligible, to over 30%), suggesting that methodological factors can play an important role. For researchers who wish to use VR for collecting movement data, this gap may mean data collected in VR, such as movement speeds and density-speed relationships, cannot reliably be used for validation of egress models or direct comparison with real-world data sets. It is useful to define this gap explicitly:

> ***Movement realism gap.*** *The systematic difference in spatio-temporal gait parameters which is observed when individuals walk overground in immersive virtual environments (VE) compared to real-world environments (RE).*

This paper addresses two obstacles to reducing the movement realism gap:
(1) the lack of unified guidance for designing VR studies to collect valid locomotion data, and
(2) incomplete reporting practices that hinder reproducibility and cross-study comparison.

Section 2 reviews the existing literature on the movement realism gap and establishes an empirical basis for the paper. We address (1) in Section 3 with experience-based design considerations, and (2) in Sections 4–5 by proposing 12 domain-specific reproducibility criteria (called W_1–W_{12} in this paper, where W denotes Walking) that complement the E/R/G categories of Zielasko and Weissker (2023) by extending reproducibility guidance to overground walking studies. Researchers designing an overground walking study in VR can use Section 3 to inform methodological design choices and Sections 4–5 for reporting guidance; reviewers may find W_1–W_{12} useful as a checklist for assessing reproducibility.

2. STATE OF THE ART: THE MOVEMENT REALISM GAP

This section reviews the existing research comparing the magnitude and direction of gait differences between walking in REs vs. VEs. In their SWOT analysis, Kinateder et al. (2014) highlighted the failure to show ecological validity as a threat to VR as a research tool. In a later publication, Kinateder et al. (2018) noted that at the time, VR could not be used to extract absolute values of locomotion parameters such as walking speed.

Research to date is limited and reported data are mixed with regard to the magnitude, and in some cases the direction, of differences in gait between walking in the RE and overground walking in VR (Sato et al., 2026). The general trend observed in the literature is that locomotion in VR results in a more 'cautious' gait pattern; namely a reduction in gait parameters of walking speed, step length, and cadence, and increases in gait parameters related to base of support and stability such as step width and double support time (Mohler et al., 2007; Yamagami et al., 2020). This cautious gait is similar to that observed in elderly populations for real-world walking (Winter et al., 1990) and is indicative of users modifying their gait to improve their stability and safety.

Across the 19 studies identified and reviewed in this current work, reported differences in walking speed between RE and VE ranged from negligible increases of +1.7% (Janeh et al., 2018) through to substantial decreases of approximately -37% (Menegoni et al., 2009). The full list of surveyed studies is provided in an online supplement (de Schot et al., 2026). Walking speed reductions in the range of 5–15% are commonly reported, although some studies using congruent environments and familiarisation procedures have achieved differences of approximately 5% or below (de Schot et al., in review; Palmisano et al., 2022). Where reported, step width tends to increase in VR, consistent with a wider base of support for stability, however this is not consistent across studies, with magnitude and direction of changes in step width between RE-VE ranging from -5% to +20%. Double support time shows increases of between 2–36% in VR, reflecting a more conservative gait. Cadence results are more variable; some studies report reductions in the order of 0.7–13.3%, while others find small increases in cadence of 0.6–2.3%. Notably, the direction of effects is not always consistent, with one study reporting faster walking speeds in VR for certain conditions. This variability across studies, and across parameters within the same study, reinforces the conclusion that the magnitude of this 'movement realism gap' is not a fixed property of VR but is influenced by methodological choices. The challenge is to identify which design decisions drive these differences and to report them with sufficient detail so that results can be compared and reproduced.

The reasons behind this movement realism gap are not well understood, however there are a number of potential causal factors, some of which can be addressed by methodology. If the objective is to use VR to collect locomotion data and to draw inferences about real world movement of individuals, there is a need to consider this movement realism gap and understand how design choices for VR scenarios can influence the validity of data collected.

3. DESIGN CONSIDERATIONS FOR VALID MOVEMENT DATA COLLECTION IN VR

This section addresses obstacle (1) – the lack of unified guidance – and sets out experience-based design considerations to help researchers make informed methodological choices when designing VR studies involving human movement analysis. Several methodological factors influence gait behaviour in VR. The factors below are identified from both the existing literature and from the authors' own VR locomotion research. The choice of experimental design depends on the specific research objectives, and some choices may be more appropriate than others depending on the research questions to be answered and any imposed constraints. In the following subsections, key methodological choices are briefly discussed to introduce their impacts on gait behaviour.

3.1 Task design

It is recognised that the nature of the task and instructions given to participants can have a strong influence on their behaviours. For example, asking someone to walk to a target may be an ambiguous instruction and result in a variety of walking speeds or styles; some participants may decide to walk carefully with perfect posture as they are aware of being recorded, others may walk casually, others may walk faster than normal. The choice of instruction wording (e.g. 'walk', 'walk at a comfortable pace', 'walk normally', 'walk as fast as you can') directly influences the observed walking speed and should be reported to allow for meaningful comparison across studies.

3.2 Spatial congruence

Spatial congruence – the degree to which the VE matches the RE in geometry and appearance – can influence gait behaviour. When participants are immersed in a VE which they know does not match their real spatial environment, they tend to be more cautious with their movement behaviour, characterised by reductions in walking speed and step length (Mousas et al., 2020). This phenomenon has been termed "collision anxiety" and is defined as "*the feeling of discomfort, anxiety or fear of colliding with or hitting the real-world environment or another person while using an XR application*" (Ring et al., 2024). Collision anxiety has been noted in multiple studies despite efforts from researchers to operate in an empty room and provide users with a virtual boundary system warning of any imminent contact with real-world obstacles (Boletsis & Cedergren, 2019; de Schot et al., 2023). Ring et al. (2024) developed a Collision Anxiety Questionnaire (CAQ) which can be used to assess a user's collision anxiety on three subscales: general collision anxiety, orientation, and interpersonal collision anxiety. Given the potential for confounding effects of collision anxiety, researchers may consider administering the CAQ, pre- and post-exposure, to determine if collision anxiety may be an important factor influencing the results.

To reduce collision anxiety, it is important to build spatial confidence and trust in the environment. This includes both the built environment (walls, obstructions), as well as the presence of other people in the shared physical environment. This can be achieved by careful spatial matching and alignment of an exact virtual replica of a space with the RE. Trust-building can also extend to the use of passive haptic effects, having users reach out and touch walls or objects which are collocated in both the VE and the RE.

3.3 Embodiment

Embodiment is a user's sense of body ownership, self-location and agency of a virtual avatar (Kilteni et al., 2012). In tasks where visual feedback from one's own body movements (visuomotor feedback) is important, the use of a tracked avatar can enhance user performance. The presence of a self-avatar has been shown to improve accuracy in distance perception (Mohler et al., 2010), improve performance in coordinated tasks like accurately walking on stepping stones (McManus et al., 2011), and even improve performance in cognitive tasks in VR (Pan & Steed, 2019). However, evidence for a direct effect of embodiment on spatio-temporal gait parameters remains limited, with recent work suggesting that avatar representation can successfully modulate sense of embodiment without producing significant changes in walking speed, cadence or step length (de Schot et al., in review). There are a number of standardised questionnaires available for researchers to measure embodiment, such as the pESQ (Eubanks et al., 2021), or the EQ (Gonzalez-Franco & Peck, 2018).

3.4 Acclimatisation to novelty

Another potential cause of differences between RE and VE locomotion is related to novelty. While VR is becoming increasingly common, many consumer applications are designed for stationary or limited movement use (presumably due to real-world obstructions in living rooms) and are not room-scale to allow for overground walking as a locomotion technique. As a consequence, even people familiar with VR may be inexperienced with overground walking through a room-scale environment. Studies have shown that gait differences between RE and VE reduce with an adaptation effect over time (Janeh et al., 2019; Martelli et al., 2019), but findings are mixed. Nevertheless, it is still considered useful to incorporate a familiarisation phase at the beginning of an experiment to allow participants to adapt to walking in VR and reduce any confounding effects of novelty.

3.5 Cybersickness

Cybersickness is a potential confounding factor in many VR user studies. It arises from the mismatch between the vestibular system (the inner-ear system that senses motion and balance) and visual systems: vection (the sensation of movement by optic flow) in VR can cause the visual system to detect movement that is inconsistent with vestibular system input. The resulting conflict can produce symptoms such as nausea and oculomotor discomfort (e.g., eye strain, focusing difficulty). To assess any influence of cybersickness on results, it is useful to administer a validated questionnaire pre- and post-exposure. Common instruments include the SSQ (Kennedy & Lane, 1993), VRSQ (H. K. Kim et al., 2018), and CSQ-VR (Kourtesis et al., 2023). The SSQ was originally developed for military flight simulator applications and is widely used in VR research, despite recent criticisms of its applicability to cybersickness measurement in modern HMDs (Kourtesis et al., 2023).

3.6 Latency

Latency has been shown to affect both general comfort and walking behaviour in VR. For walking in VR, the most relevant metric is motion-to-photon (MTP) latency; defined as the time delay between a movement of a tracked object, and the corresponding visual update rendered in the user's display (Stauffert et al., 2020). MTP latency is an inherent system-level property of a VR system's architecture and includes latencies within the tracking system, network, processing pipeline, and rendering stage (Figure 1), hence reported MTP latency should be measured rather than assuming values from manufacturer's specifications. Increased latency contributes to cybersickness, with threshold values varying by the nature of the application (J. Kim et al., 2020). Additional latency can also reduce sense of embodiment (Samaraweera et al., 2015), with system latencies being highly dependent on VE design choices and rendering load (Feldstein & Ellis, 2021).

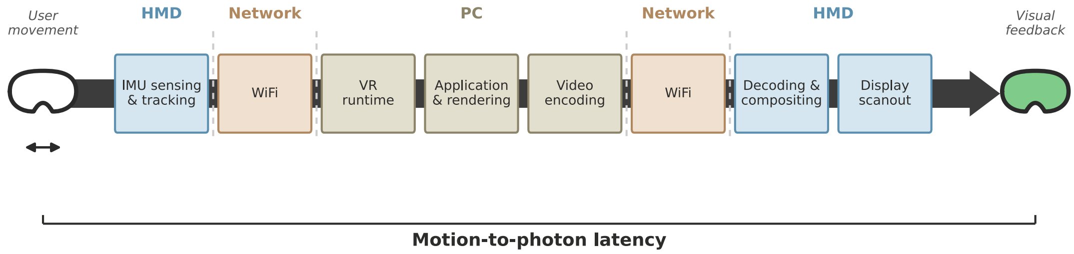

Figure 1. Overview of latency sources in a PC-based VR configuration.

3.7 Validity thresholds

A question that arises when considering the validity of data is, 'how close is close enough?'. The answer depends on the application: a microscopic gait-based evacuation model may demand centimetre-resolution agreement in gait parameters, whereas wayfinding or exit choice studies may prioritise behavioural validity and tolerate a larger movement realism gap. Conventional null hypothesis significance testing is poorly suited to this question, as a failure to reject a null hypothesis of no difference is not the same as demonstrating equivalence. A more appropriate method is equivalence testing, e.g., via the two one-sided tests (TOST) procedure (Lakens, 2017); the principal challenge here is setting the equivalence margin.

A useful concept borrowed from biomechanical and clinical research is the Minimal Clinically Important Difference (MCID). For the parameter of walking speed, the MCID has been described as approximately 0.1 m/s (Bohannon & Glenney, 2014); differences within this threshold are not considered clinically relevant. However, MCID (or MID) values are anchored to specific clinical populations and may vary by population and context (King, 2011); their transposition into healthy populations lacks a clear basis. A proportionate approach (for example ±10% of each participant's RE baseline) avoids this issue, scales appropriately across a range of speeds, and at typical comfortable walking speeds (≈1.1 m/s) remains in approximate alignment with the lower end of the MCID range. Researchers should consider defining and reporting equivalence margins appropriate to their specific application and population, rather than relying on either significance testing alone or borrowed clinical thresholds.

4. LITERATURE REVIEW OF VR LOCOMOTION STUDIES USING OVERGROUND WALKING

With the aforementioned design considerations in mind and building on the approach by Zielasko and Weissker (2023), we developed 12 domain-specific reporting criteria (W_1–W_{12}) for VR locomotion studies using overground walking as the locomotion technique, and applied them to assess reporting completeness across a range of published research addressing obstacle (2) – incomplete reporting practices.

4.1 Method

We collected 19 studies, identified through citation chaining and domain expertise, that compared spatio-temporal gait parameters between RE and VE during overground walking. This approach captured research where RE-VE gait comparison was secondary to the primary research objectives which may have been missed during a formal database search. This selection of studies is not exhaustive, and selection did not follow a systematic literature search process; however, it represents an accessible evidence base that informed the development of the proposed reporting guidelines.

In a similar method to Zielasko and Weissker (2023), we reviewed the selected papers and extracted information on a series of criteria that we considered important to fully understand the findings, but also necessary to be able to reproduce the results. To evaluate reporting completeness, 12 attributes were defined (W_1–W_{12}; Section 5) with operational criteria for coding. The selected criteria are organised across four areas: **Walking task** (W_1: walking path dimensions and layout; W_2: familiarisation procedures; W_3: task instructions), **Measurement** (W_4: explicit operational definitions for each gait parameter; W_5: gait event detection criteria; W_6: measurement system, calibration, and accuracy), **Technical Setup** (W_7: HMD specifications; W_8: hardware and software environment; W_9: self-avatar representation), and **Environment & Participants** (W_{10}: VE-RE spatial relationship; W_{11}: prior VR experience quantitatively; W_{12}: ethics approval). Each was coded as fully reported (Y), partially reported (P), or not reported (N). A criterion was coded as fully reported where there was sufficient detail provided to enable replication; partially reported where information was present but was ambiguous or incomplete; and not reported where the information was absent.

4.2 Results

Results revealed substantial gaps in reporting (Figure 2). While most studies adequately reported path geometry (74%), walking instructions (79%) and software environment (84%), several important criteria showed poor reporting rates: measurement system description and accuracy (32%), avatar representation (37%), prior VR experience (42%), and familiarisation procedures (53%). Notably, two of the worst reported criteria, avatar representation and prior VR experience, correspond with methodological factors identified in Section 3 as potential contributors to the movement realism gap. The third, measurement system accuracy, is important to be able to interpret the magnitude of any reported differences. Together, these gaps limit reproducibility and make it difficult to determine whether variability between studies reflects genuine effects or methodological differences.

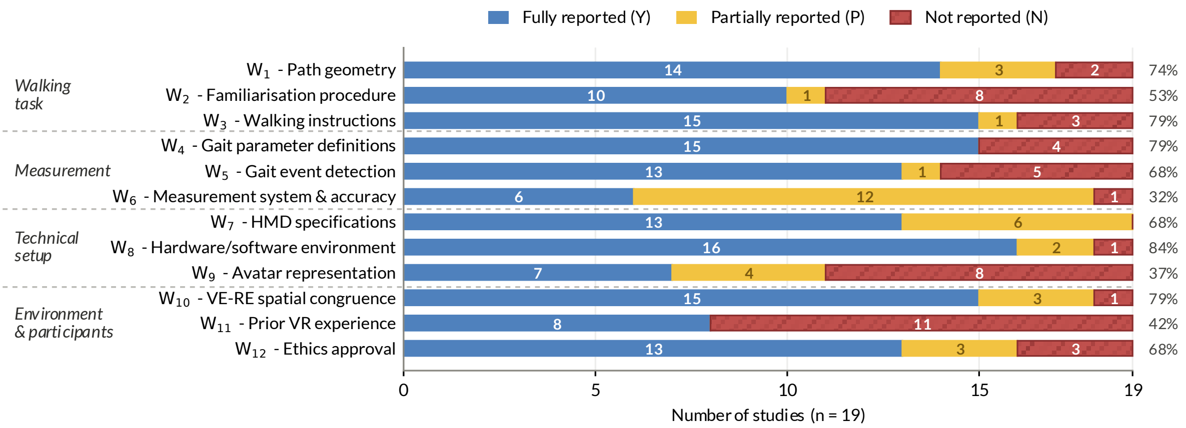

Figure 2. Completeness of reporting for the 19 surveyed studies. Y = Fully reported, P = Partially reported (ambiguous or incomplete), N = Not reported. Within-bar values are counts; right margin values show the percentage of studies fully reporting each criterion.

5. REPRODUCIBILITY GUIDELINES

This section presents recommendations for reporting guidelines to aid reproducibility and cross-study comparison of findings from VR overground walking studies. The guidelines are organised into the four areas introduced in Section 4; Walking task, Measurement, Technical setup, and Environment & participants. Each criterion is presented as a definition of what to report, followed by an extract from a VR locomotion paper (de Schot et al., in review) as an

example. Several criteria correspond directly to design considerations introduced in Section 3 and are cross-referenced accordingly. Although these criteria are presented in a VR context, several of them apply equally to RE locomotion studies (W_1 – Path geometry, W_3 – Walking instructions, W_4 – Gait parameter definitions, W_5 – Gait event detection, W_6 – Measurement system accuracy, and W_{12} – Ethics approval).
In addition to the domain-specific criteria below, researchers will also find it useful to follow the general reproducibility guidelines established by Zielasko and Weissker (2023), which include comprehensive guidance on the descriptions of experimental setup (E_1–E_5), sample population reporting (E_6–E_{10}), guidance on enhancing scientific rigour (R_1–R_6), and descriptions on the generalisability of results (G_1–G_3). The W_1–W_{12} criteria presented in this paper are intended to complement, not replace, those general guidelines by addressing the specific case of overground walking studies, which was excluded in the work of Zielasko and Weissker (2023).

5.1 Walking task

W_1 – *Path geometry*. Report walking path dimensions, shape, and layout of the real environment, ideally supported with a diagram.
Example: The walking task used a typical one-dimensional loop walking configuration measuring 0.8 m in width, with straight portions of the loop measuring 4.0 m in length. Gait events were analysed within a measuring zone spanning the full length and width of this section (x = 0 to 4.0 m, y = 0 to 0.8 m in the Motive coordinate system; see Figure 3 for dimensions and the location of the Motive coordinate system origin). Extract from de Schot et al. (in review).

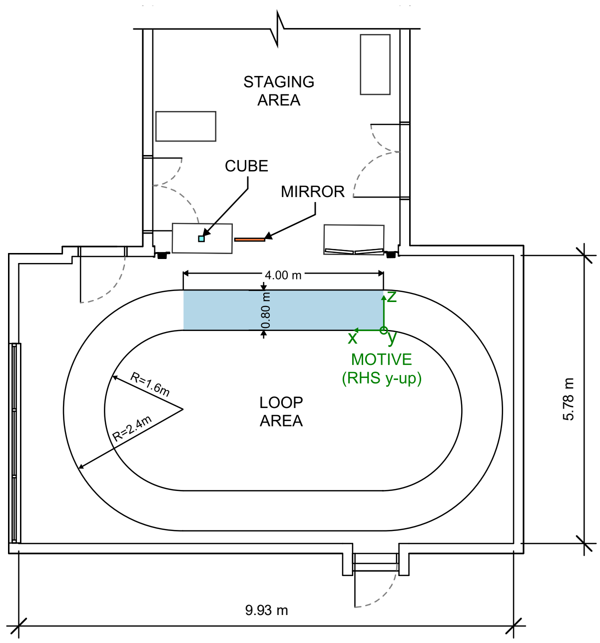

Figure 3. Geometry and dimensions of the experimental setup.

W_2 – *Familiarisation procedure*. Describe any acclimatisation, practice, or trust-building procedures conducted before data collection (see Section 3.4).
Example: Once all equipment was fitted, a familiarisation phase was conducted where participants were placed in the virtual environment (with no avatar). They were then given an explanation that the VE was a replica of the actual laboratory, being of similar appearance and the same size and location as the real space. Participants were asked to pay attention to the markings on the walls and floor, particularly the yellow loop walking track which they would soon be walking along. They were then asked to look around and identify a yellow square and once found to walk over and reach out and touch the square, thus providing a passive haptic effect and confirming the co-location of the VE and RE. Participants were allowed between 60–120 s for this phase. Extract from de Schot et al. (in review).

W_3 – *Walking instructions*. Report any task-specific instructions given to participants, such as pace guidance ‘walk at a comfortable pace’, behavioural constraints ‘don’t overtake’, or directional cues ‘walk in a counter-clockwise direction’ (see Section 3.1).
Example: Immediately following the body awareness tasks, the walking task was instructed by audio cue: ‘For the next task you will walk around the loop three times. You will first hear a countdown, when this finishes and you hear the word “Go”, please start walking around the loop at a comfortable pace until you are told to stop walking.’ Extract from de Schot et al. (in review).

5.2 Measurement

W_4 – *Gait parameter definitions*. Provide explicit operational definitions for each reported gait parameter (e.g., mean walking speed defined as distance divided by time between

specified markers). It is useful to align with existing naming conventions and definitions found in the gait and biomechanics research community.
Example: A gait cycle from right heel strike is shown in Figure 4 [in de Schot et al. (in review)] which also contains illustrative definition of the spatio-temporal gait parameters selected for this study, chosen for comparison with common measures reported in existing literature:

- Walking speed - the mean speed during traversal of the 4 m measuring zone.
- Cadence - the number of steps taken per minute during the measuring zone.
- Double Support % - the amount of time (expressed as a percentage of the gait cycle) that a participant spends with both feet in contact with the ground.
- Step length - the anterior-posterior distance between heel markers at successive heel strikes.
- Step width - the mediolateral distance between left and right feet. Calculated as the perpendicular distance between the line connecting the two ipsilateral foot heel points and the contralateral heel point to align with adjacent research on VR gait. Extract from de Schot et al. (in review).

W$_5$ – *Gait event detection*. Specify the criteria used to detect gait events, such as heel strike or toe off (e.g., velocity threshold, vertical displacement minimum, force plate trigger). This could be as simple as when a tracked marker passes a spatial threshold in a walking track.
Example: Heel strikes were identified as local minima in vertical heel displacement of the LHeel and Rheel markers, with smoothing and prominence thresholds tuned for accurate detection. Toe off events were similarly detected using local minima in the vertical displacement of LToeIn and RToeIn markers. Walking speed detection events occurred as the WaistBack marker crossed the x = 0 m and x = 4 m boundaries of the measuring zone. Extract from de Schot et al. (in review).

W$_6$ – *Measurement system accuracy*. Describe the measuring system. This should include the tracking technology (e.g., optical motion capture, overhead camera video-based tracking, HMD-based tracking), sampling rate, system accuracy, and any reported calibration residuals or measurement error bounds.
Example: To collect participants' kinematic data, an OptiTrack motion capture (Mocap) system was used which comprised ten OptiTrack cameras (4 x Prime 13, 4 x Prime 13x, and 2 x Prime 13W) connected through a D-Link DGS-1250-28XMP PoE switch to the lab PC which was running the Motive software (version 2.3.3). Camera frame rates were set to 200 Hz. Participants had the Helen Hayes lower body marker set with 14 mm diameter retro-reflective passive markers. OptiTrack calibration was confirmed daily within Motive to ensure errors were less than 0.3 mm/ray. Extract from de Schot et al. (in review).

5.3 Technical setup

W$_7$ – *HMD specifications*. Report the HMD specifications. At a minimum this should include the make/model, field of view, resolution, tethered/wireless, refresh rate. This should also include reported or measured motion-to-photon latency (see Section 3.6), noting that this is a system-level property influenced by rendering pipeline and tracking subsystem in addition to the HMD hardware.
Example: The VR system comprised a VIVE Focus 3 Head-Mounted Display (HMD) which was used in PCVR mode with VIVE Business Streaming (version 2.1.5a) over a local WiFi-6 network. The Focus 3 weighs approximately 808 g with VIVE Ultimate Tracker dongle, has an IPD range of 57–72 mm, a resolution of 2448 x 2448 pixels per eye, a refresh rate of 90 Hz, and FoV of 116° horizontal and 96° vertical. MTP latency was measured with an iPhone 14 Pro at 240 fps using a frame counting method with a custom script to change HMD display to red when rotation was detected. Extract from de Schot et al. (in review).

W$_8$ – *Hardware/Software environment*. Describe the hardware and software. This includes aspects such as PC hardware (CPU, GPU) and software used for the model development, and VR runtime software, rendering software, virtual environment model specifications, and any relevant display settings.
Example: Steam VR (version 2.9.6) was used to run the experimental scenario directly from the Unreal Engine 5.3 (UE5.3) editor. The VR scenario was run on a PC with an AMD Ryzen 9 7900X CPU, NVIDIA RTX 4080 GPU, 64 GB RAM, and Windows 10. Extract from de Schot et al. (in review).

W$_9$ **–** *Avatar representation*. At a minimum, the presence (or absence) of any self-avatar representation should be included. If there is self-avatar representation, the tracking method, and appearance of visible components should be specified. Images of the self-avatar are useful for the reader (see Section 3.3).
Example: The conditions varied between a low embodiment condition (VE1) where there was only a virtual camera and no self-avatar, a medium embodiment condition (VE2) where a user had a static self-avatar (where the avatar translated and rotated with the HMD, but did not incorporate any tracking of limbs), and a high embodiment condition (VE3) where the user had a self-avatar with full-body tracking (FBT). The avatar selected was an androgynous android 'Emmanuel' from the Existent® library. Extract from de Schot et al. (in review).

W$_{10}$ **–** *VE-RE spatial congruence*. Describe the spatial relationship between the virtual and physical environments including whether geometry was matched, similar, or different, whether visual appearance was replicated and to what level of fidelity (see Section 3.2). Side by side images of the real and virtual environments are useful for the reader.
Example: A virtual replica of the physical laboratory was created. Care was taken to ensure the spatial and visual congruence of the VE and RE. A 3D model of the physical lab was created in Revit with room dimensions taken using a laser measuring device. This Revit model was imported into UE5.3 using the Datasmith pipeline. This lab model was not photo-realistic but included details such as coloured tape tracking markings on the floors and walls, and the loop walking track. See Figure 4 for views of the RE and VE versions of the laboratory. Extract from de Schot et al. (in review).

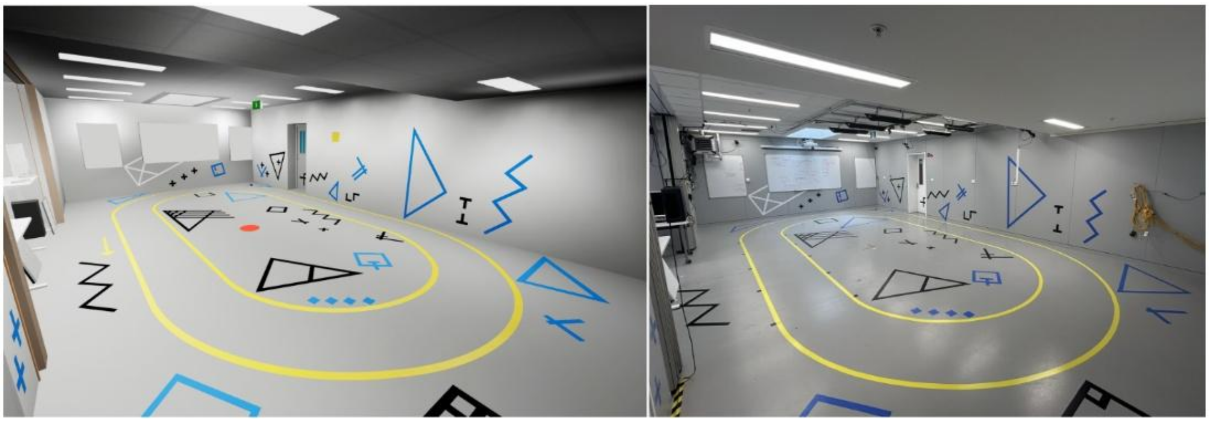

Figure 4. (Left) virtual replica of the lab used in VE conditions, (right) the RE laboratory. de Schot et al. (in review).

5.4 Environment & participants

W$_{11}$ **–** *Prior VR experience*. Report participants' prior exposure to or experience with VR. Rather than a Boolean (Y/N) response, consider collecting self-reported competence, frequency of use (daily, weekly, monthly), cumulative duration (hours), and tenure (time since first VR exposure).
Example: Prior VR experience was assessed via a post-experiment questionnaire capturing frequency of use, tenure, and self-reported sense of VR competence (see supplementary materials). Responses were operationalised into a summed score ranging between 3–15. Extract from de Schot et al. (in review).

W$_{12}$ **–** *Ethics approval*. While not strictly a methodological criterion, ethical review and approval is included given the physical and psychological risks inherent in research involving human participants and the importance of ethical oversight to the credibility of findings. Report institutional ethics review and approval including committee reference number. Other useful information includes the use and amount of any remuneration or enticement.
Example: The project was subject to review and approved by the University of Canterbury Human Ethics Committee (reference HEC 2020/94/LR Amdt. 4). Participants provided informed consent. A movie ticket was provided as reimbursement; participants were advised they could end the experiment at any time of their

choosing and still receive this reimbursement. However, none of the participants terminated the experiment. Extract from de Schot et al. (in review).

6. LIMITATIONS AND FUTURE WORK

Evacuation may involve other gait modes such as running, stair descent/ascent, crawling, and wading; the present work examines level-ground walking only. Research validating gait data for these other modes in VR remains limited, making level-ground walking a logical starting point. Similarly, there are many VR locomotion techniques, such as repositioning devices (e.g. treadmills), or redirected walking; the present work focused only on overground (or 'real') walking, so findings may not directly translate to other VR locomotion techniques. Nevertheless, the findings from this paper are expected to be informative for researchers exploring these other evacuation gait modes and for those working with other VR locomotion techniques.

The 19 studies reviewed were identified through citation chaining and domain expertise rather than a formal systematic search. Consequently, the sample may not be exhaustive. Additionally, reporting completeness was coded by a single reviewer without inter-rater reliability assessment, which may introduce subjectivity in the distinction between fully and partially reported criteria. Future work could incorporate inter-rater reliability and extend the W_1–W_{12} criteria to other VR locomotion techniques.

7. CONCLUSIONS

Our review of 19 studies reveals substantial gaps in reporting, particularly for measurement accuracy, avatar representation, and prior VR experience. These gaps limit reproducibility and cross-study comparison, making it difficult to determine which methodological choices influence the movement realism gap. The W_1–W_{12} criteria presented in this paper provide domain-specific reporting guidelines for researchers, enabling systematic comparisons across studies and supporting understanding of when VR can reliably capture human locomotion. We invite authors to use W_1–W_{12} as a self-assessment checklist during experimental design and manuscript preparation; reviewers may also find them useful to evaluate reporting completeness during peer review.

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